

Ali Raza

Full-Stack AI Engineer

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PROFESSIONAL SUMMARY

Full-Stack AI Engineer and Software Engineering undergraduate building intelligent products across React/TypeScript and Next.js frontends, FastAPI services, PostgreSQL/Redis data layers, LLM and agentic workflows, real-time voice AI agents, and n8n / Make.com / GoHighLevel automation pipelines. Designed a containerized AI-powered workspace platform with secure authentication, role-aware access, real-time WebSockets, and Groq-powered insights, architected a reusable GoHighLevel CRM snapshot that deploys a full lead-to-close system for real estate clients in under an hour, and built a multilingual hotel receptionist voice agent that books rooms and confirms them on WhatsApp. Further experience spans document and video RAG (Self-RAG and CRAG patterns), LangGraph agents, LangSmith tracing, Flutter mobile development, NLP, and computer vision, with a focus on turning AI prototypes into complete, usable software systems.

TECHNICAL SKILLS

Full-Stack & Platform Engineering: Python, FastAPI, React, Next.js, TypeScript, JavaScript, Vite, Tailwind CSS, Flutter, Dart, REST APIs, WebSockets, JWT/OAuth2, role-aware authorization, Pydantic

Generative AI & LLMs: RAG pipelines, Self-RAG, CRAG, agentic systems, prompt engineering, function calling, embeddings, semantic search, MMR retrieval, LangChain, LangGraph, LangSmith, ReAct, Groq API, Anthropic and OpenAI models, Hugging Face, Llama models

Voice AI: LiveKit Agents, Silero VAD, Whisper speech-to-text (Groq), ElevenLabs multilingual TTS, real-time conversational agents, multilingual language mirroring

Data, Persistence & Background Processing: PostgreSQL, PGVector, Redis, Celery, Tortoise ORM, Aerich, SQLAlchemy, SQLite

Machine Learning, NLP & Vision: Scikit-learn, TensorFlow, PyTorch, Pandas, NumPy, TF-IDF, sentiment analysis, classification, clustering, SVM, Random Forest, Decision Trees, K-Means, Naive Bayes, regression, CNNs, LSTMs, transfer learning, ResNet50

Workflow Automation & Deployment: n8n, Make.com, GoHighLevel (CRM architecture, funnels, workflows, Conversation AI, A2P 10DLC compliance), Evolution API (WhatsApp), Docker Compose, Vercel, Streamlit, Git, GitHub, Jupyter Notebook, Google Colab, Kaggle, OpenAPI/Swagger

SELECTED PROJECTS

Reusable GoHighLevel Real Estate CRM & Automation Snapshot [Docs]

GoHighLevel, Conversation AI, HTML email, A2P 10DLC

- Architected a client-agnostic GoHighLevel snapshot covering the full lead-to-close lifecycle: 14 workflows, 53 custom fields, 41 tags, 3 pipelines, 4 calendars, 6 landing pages, and 32 SMS/email templates.
- Built deployment around 30 Custom Values so every asset rebrands from a single configuration table, cutting per-client onboarding from a day of manual edits to under an hour.
- Configured a Conversation AI agent for inbound qualification and calendar booking with defined escalation rules, alongside sub-60-second speed-to-lead, lead scoring, no-show recovery, and 45-day buyer/seller nurture sequences.
- Engineered compliance into every workflow: A2P 10DLC-ready consent language, DND checks before every send, and exit conditions on all automated sequences.

Multilingual AI Hotel Receptionist Voice Agent

Python, LiveKit Agents, Groq, ElevenLabs, Silero VAD, n8n, Google Sheets, Evolution API

- Built a real-time voice receptionist on LiveKit Agents with Silero VAD, Groq Whisper-large-v3-turbo transcription, a Groq-hosted LLM, and ElevenLabs multilingual TTS, detecting the caller's language and replying in it (Urdu, English, Spanish).
- Implemented LLM function calling for room booking that collects guest name, room type, and phone number, then posts to an n8n webhook with timeout and error handling so the agent recovers gracefully on failure.
- Orchestrated n8n workflows that log every reservation to Google Sheets and send instant WhatsApp confirmations through a self-hosted Evolution API, plus a room-availability webhook returning status and pricing to the agent.

Intelligent AI Assistant & Content Automation Pipelines [Code]

n8n, Make.com, Groq, LangChain, Google Workspace APIs, Meta API

- Engineered a Personal Assistant Agent within n8n, using Groq and LangChain to automate Google Calendar scheduling, Gmail management, Google Docs generation, and Google Sheets expense tracking.
- Developed a modular content distribution engine polling Google Sheets for state management and auto-publishing media to YouTube and Instagram Reels via the Meta Graph API.
- Implemented asynchronous branching, parallel execution tracks, and web search integration for multi-step agents.

AI-Powered Workspace Management Platform [Live | Code]

Python, FastAPI, React, PostgreSQL, Redis, Celery, Docker

- Designed a full-stack collaboration platform for workspaces, teams, projects, tasks, files, and analytics.
- Developed secure JWT authentication, role-aware authorization, real-time WebSockets, and Groq-powered insights.
- Added Redis caching, Celery background reports, and CSV exports; containerized services with Docker Compose.

AI Chat Assistant (ChatGPT Clone) [Live | Code]

React, TypeScript, FastAPI, LangGraph, PostgreSQL, Docker

- Built a production-style assistant with React/TypeScript, FastAPI, and LangGraph serving Groq-hosted gpt-oss-120b.
- Implemented token-by-token streaming, persistent conversations, stop/regenerate controls, and Markdown rendering.

IUB University & YouTube RAG Chatbots [Code]

LangChain, PGVector, PostgreSQL, Hugging Face, Groq, Streamlit

- Built RAG assistants to ingest official IUB PDFs and YouTube transcripts (via yt-dlp), storing vectors in PGVector.
- Orchestrated grounded answer generation with Llama 3.1 via Groq and MMR retrieval behind Streamlit interfaces.

Additional Projects [Portfolio | Repos]

Next.js, Tailwind CSS, LangGraph, FastAPI, Tortoise ORM, Scikit-learn, TensorFlow, PyTorch

- Deployed a personal engineering portfolio featuring *RazaMind AI*, an embedded assistant answering visitor questions about projects; built a multi-tool ReAct agent and a reusable FastAPI/Tortoise ORM authentication service.
- Published 10+ end-to-end ML/DL projects: CIFAR-10 with ResNet50, face mask detection, cat-vs-dog and Fashion-MNIST CNNs, MNIST digits, breast cancer and heart disease prediction, fake news detection, sentiment analysis, credit card fraud detection, and Titanic survival modelling.

PROFESSIONAL EXPERIENCE

Machine Learning Engineer Intern | Developers Hub Corporation (Remote)

2025

- Built an end-to-end IMDB sentiment system using TF-IDF features, linear classifiers, and a Streamlit interface; developed ML workflows for NLP, classification, and clustering with reproducible prediction pipelines.

Machine Learning Engineer Intern | Code Alpha (Remote)

2025

- Delivered projects in heart-disease prediction, agricultural image recognition, and speech-emotion recognition, training logistic-regression, CNN, and LSTM models with structured preprocessing and performance analysis.

EDUCATION

BS Software Engineering | The Islamia University of Bahawalpur

2023-2027

FSc Pre-Engineering | KIPS College

2021-2023

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- Automation & CRM:** Delivered a production-ready GoHighLevel (GHL) snapshot system covering CRM structure, funnels, workflows, Conversation AI, and compliance; completed n8n and Make.com training for agentic AI and business automation pipelines.
- Generative AI & LLM Engineering:** Completed CampusX coursework in LangChain, RAG pipelines, and ReAct agents.
- ML/DL & Backend Engineering:** Completed Hope to Skill and Siddhardhan ML/DL tracks, plus FastAPI, Tortoise ORM, JWT, PostgreSQL, Redis, Celery, and Docker training.
- One Million Prompters:** Dubai Future Foundation and Dubai Centre for Artificial Intelligence.